$\boxed{\text{Case 4:}}$ a is odd, b is odd. By Defn. 1, $\exists$ integers K_1 and K_2 such that

$a = 2K_1+1$ and $b = 2K_2+1$

$$a^3+ab^2+b^3 = (2K_1+1)^3+(2K_1+1)(2K_2+1)^2+(2K_2+1)^3$$

$$= (8K_1^3+12K_1^2+6K_1+1)+(2K_1+1)(4K_2^2+4K_2+1)+(8K_2^3+12K_2^2+6K_2+1)$$

$$= (8K_1^3+12K_1^2+6K_1+1)+(8K_1K_2^2+8K_1K_2+2K_1+4K_2^2+4K_2+1)+(8K_2^3+12K_2^2+6K_2+1)$$

$$= 2(4K_1^3+6K_1^2+3K_1+4K_1K_2^2+4K_1K_2+K_1+2K_2^2+2K_2+4K_2^3+6K_2^2+3K_2+1)+1$$

By Defn 1, $a^3+ab^2+b^3$ is an odd integer which contradicts the parity of RHS.

The only possibility: $a = 2K_1$ and $b = 2K_2$ $\therefore \frac{a}{b} = \frac{2K_1}{2K_2} = \frac{K_1}{K_2}$

2 being a common factor of a and b contradicts the assumption that r is in lowest terms.

$\boxed{\therefore \neg q \text{ is False.}}$ By defn. of AND, $(p \wedge \neg q)$ is False. By defn. of conditional, $(p \wedge \neg q) \rightarrow F$ is True. $\therefore p \rightarrow q$ is True.

$\boxed{\therefore \text{if } r \in R \text{ and } r^3+r+1=0, \text{ then } r \text{ is irrational}}$ $\square$

28) Prove that if n is a positive ~~po~~ integer, then n is even if and only if $7n+4$ is even.

$\boxed{\text{Soln:}}$ p: n is even $\quad q$: $7n+4$ is even

We shall prove $p \leftrightarrow q$ is True. From the tautology $(p \leftrightarrow q) \leftrightarrow ((p \rightarrow q) \wedge (q \rightarrow p))$, we shall prove $(p \rightarrow q) \wedge (q \rightarrow p)$ is True. By the defn. of AND, we need to prove both $p \rightarrow q$ and $q \rightarrow p$ are True.

$\boxed{\text{Direction 1:}}$ We shall prove $p \rightarrow q$ is True by a direct proof

Let p be True. $\therefore n$ is an even no. By defn 1, $\exists$ integer K such that

$n = 2K \quad 7n+4 = 7(2K)+4 = 2(7K+2)$